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Strategically timed UAS data acquisition for wheat yield prediction using explainable AI: A multi-year phenology-aligned study

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ABSTRACT

Accelerating genetic gain in wheat requires scalable, non-destructive high-throughput phenotyping. This study evaluated phenology-aligned UAS vegetation indices for yield prediction across the 2018–2021 growing seasons at Bushland, Texas, under dryland and irrigated conditions. Ten flights per year provided Normalized Difference Vegetation Index (NDVI), Normalized Difference Red Edge Index (NDRE), and Excess Green (ExG) measurements, synchronized using Days Relative to Heading (DRH). XGBoost, Ridge Regression, Elastic Net, Random Forest, Extra Trees, Partial Least Squares Regression, and neural networks were evaluated using repeated random splits and independent-year validation. Model-specific feature importance was assessed over 100 iterations and independently evaluated using model-agnostic permutation importance to identify consistent temporal patterns. Extra Trees and XGBoost achieved the strongest repeated random-split performance, with RMSE values of 6.12 and 6.13 bu ac^{-1}, respectively, and $R^2 = 0.953$ for both models. Leave-One-Year-Out validation revealed substantial interannual variation in prediction accuracy, highlighting environmental influences on generalization. Temporal correlation and explainability analyses identified a heading-centered sensing window, approximately −15 to 0 DRH, with informative features extending into the post-heading period. This pattern is physiologically consistent with canopy development, chlorophyll dynamics, biomass accumulation, and reproductive sink formation influencing grain yield. Within the evaluated environments, phenology alignment and complementary explainability provide a biologically interpretable framework for prioritizing UAS acquisitions. By aligning measurements to a routinely recorded breeding trait, the framework also facilitates practical implementation across breeding populations with differing phenology. The identified window offers a candidate strategy for reducing redundant flights and supporting efficient high-throughput selection, although reduced-flight performance requires direct validation.

1. Introduction

High-throughput phenotyping (HTP) has become an important component of modern plant breeding because it enables rapid, repeatable, and non-destructive characterization of crop performance across large field trials [1,2]. Among available HTP platforms, unoccupied aerial systems (UAS) have emerged as particularly valuable tools due to their operational flexibility, high spatial resolution, and ability to collect temporally repeated observations throughout the growing season [3–5]. In wheat breeding, UAS-based sensing has expanded the capacity to monitor canopy development, vigor, greenness, and structural traits across large nurseries while reducing the labor, time, and subjectivity associated with conventional field assessments [6–8].

Grain yield remains the primary target trait in wheat improvement

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programs, but it is also one of the most difficult traits to predict because it reflects the combined effects of genotype, environment, and management over time [9–12]. This complexity has motivated the use of machine learning (ML) approaches, which are well suited to modeling nonlinear relationships and high-dimensional interactions among spectral, agronomic, and environmental variables [13,14]. In UAS-based studies, vegetation indices (VIs) such as normalized difference vegetation index (NDVI), normalized difference red-edge index (NDRE), and excess green (ExG) have shown strong potential for characterizing canopy condition and serving as predictors of final yield [15–18]. Multi-temporal UAS observations are particularly valuable because they capture crop responses across developmental stages rather than providing only a single static representation of crop status [19].

Despite these advances, most UAS-based yield prediction studies have emphasized predictive accuracy more than interpretability and biological explanation [20,21]. As a result, many ML models function as black boxes, offering limited insight into which features drive predictions or which crop growth stages provide the most informative signals for yield estimation. This limits their practical value in breeding and precision agriculture, where researchers and decision-makers need not only accurate predictions, but also actionable guidance on how and when data should be collected [22,23]. In multi-temporal phenotyping campaigns, this issue is especially important because frequent UAS flights may produce redundant observations that increase operational cost and data-processing burden without proportional gains in predictive performance [4,24].

Kheir et al. [25] present a novel hybrid approach combining process-based modeling, remote sensing, and machine learning for high-resolution spatial prediction of wheat yield and quality parameters. The study demonstrates how integrating these diverse technologies significantly enhances accuracy over traditional single-model agricultural predictions. Kheir et al. [26] developed an automated machine learning (AutoML) fusion approach combining remote sensing, soil, and weather data to accurately predict smallholder wheat yields at scale in Egypt. Their stacked ensemble framework demonstrated strong performance for both early remote sensing yield forecasting and mid-century climate change impact assessments.

Explainable artificial intelligence (XAI) offers a framework for addressing this limitation by quantifying feature contributions and translating model behavior into interpretable agronomic insight [27,28]. In agricultural applications, XAI can help identify not only which predictors are important, but also when during crop development the most informative signals occur [29–32]. For time-series UAS data, this temporal interpretability is especially valuable because it enables the identification of growth-stage-specific sensing windows that contribute most strongly to yield prediction. Such information can support more efficient and biologically informed phenotyping strategies, improve trust in model outputs, and enhance the operational utility of predictive tools in smart agricultural systems [15,24,27].

A further challenge in multi-year UAS studies is the temporal misalignment of crop development across seasons. Differences in planting date, heading date, and environmental conditions can cause the same calendar date to represent different physiological stages in different years, thereby weakening the consistency of temporal comparisons. Days after planting is not a biologically meaningful metric for winter wheat because year-to-year differences in planting date can misrepresent crop developmental stage [5]. In wheat, heading is a biologically meaningful developmental milestone that marks the transition from late vegetative growth to reproductive development and is closely associated with processes influencing final grain yield [33,34]. Aligning observations using days relative to heading (DRH), calculated as flight date minus heading date, provides a phenology-based framework for standardizing UAS measurements across years and environments. This approach links spectral observations to comparable physiological stages rather than fixed calendar dates, thereby improving the biological consistency and transferability of temporal yield-prediction models.

Despite these advances, existing UAS-based wheat yield prediction studies have generally focused on improving predictive accuracy or identifying important predictor variables, with comparatively less emphasis on integrating phenology-aligned temporal analysis and explainable AI to support biologically meaningful interpretation and operational decision-making. Although previous studies have demonstrated the value of UAS imagery and machine learning for yield prediction, relatively few have systematically combined multi-year phenology alignment with explainable AI to identify temporally stable and operationally efficient sensing windows for wheat breeding applications.

While prior studies have investigated UAS remote sensing, machine learning modeling, and explainable AI in isolation, existing literature predominantly relies on fixed calendar dates or Days After Planting (DAP) for spectral feature extraction, an approach highly vulnerable to inter-annual climatic shifts and genotypic variations in phenology. One of the primary methodological contributions of this study lies in establishing a phenologically synchronized, physiological-time framework (centered on Days Relative to Heading, DRH) combined with multi-model feature importance quantification to resolve cross-year prediction instability. Unlike standard empirical models that treat UAS data as static snapshot inputs, our integrated pipeline explicitly aligns multispectral indices across critical developmental windows, transforming black-box ensemble predictions into physiologically interpretable yield drivers. Furthermore, by systematically evaluating model performance against out-of-year validation datasets, we demonstrate that phenology-driven alignment reduces cross-year prediction error, achieving a minimum RMSE of 6.03 bu ac^{-1} with tree-based ensembles, while defining a minimized, cost-effective sensing window ($-$ 15 to 0 DRH) that significantly outperforms conventional fixed-date approaches [15,23]. Consequently, the translation of model interpretation into practical guidance for optimizing UAS data acquisition remains limited. Addressing this gap requires predictive frameworks that are not only accurate but also interpretable, biologically grounded, and capable of guiding efficient data acquisition under practical resource constraints. Furthermore, robust interpretation requires explainability methods that provide consistent and reliable estimates of feature importance across different machine learning algorithms. Integrating model-specific explainability with independent model-agnostic validation offers greater confidence that identified temporal patterns represent genuine biological relationships rather than artifacts of a particular interpretation method. Therefore, the objectives of this study were to: (i) evaluate the predictive performance of multiple machine learning models for wheat grain yield using multi-year UAS-derived vegetation indices; (ii) develop a phenology-aligned explainable AI framework to quantify the temporal importance across crop developmental stages; and (iii) identify biologically interpretable and operationally efficient UAS acquisition windows for wheat yield prediction. This study introduces an integrated analytical framework that combines phenology alignment, multi-model machine learning, model-specific explainable AI, and model-agnostic permutation importance validation to translate model interpretation into operational recommendations for UAS data acquisition, thereby moving beyond black-box prediction toward robust, transparent, and actionable decision support for high-throughput wheat breeding and smart agricultural monitoring.

2. Materials and methods

2.1. Study area

Winter wheat advanced yield trials were conducted from 2017–2021 under both dryland and sprinkler-irrigated conditions at the Texas A&M AgriLife Research Experiment Station at Bushland, TX (35°11′ N, 102°06′ W, elevation 1170 m). Trials were established annually in October using a randomized complete block design with two to three replications depending on the stage of the breeding pipeline.

Experimental units consisted of small plots measuring 4.5 m × 1.5 m under irrigation and 6.1 m × 1.5 m under dryland conditions, each planted with seven rows spaced 0.18 m apart. Trials represented Years 7–12 of the breeding cycle, encompassing preliminary through advanced yield testing stages (Table 1). Standard agronomic practices for the Texas High Plains were followed, including fertility and weed management as required (Table 2). Nine semi-permanent 0.3 m × 0.3 m ground control points (GCPs) painted in a black and white checkboard pattern were placed across the field after planting, and precise coordinates of the GCPs were surveyed using Trimble R8s Global Positioning System (GPS; Trimble Inc., Sunnyvale, CA, USA) by accessing TXDOT's RTK network.

2.2. UAS data collection

UAS-data collection occurred throughout the eight-month growing season each year using a DJI Phantom 4 Pro equipped with an integrated RGB sensor and DJI Matrice 100 equipped with SlantRange 3P multispectral sensor (SlantRange, Inc., San Diego, CA, USA). The specifications of each sensor are detailed in Table 3. Across years, 20-32 flights were conducted per season per environment, capturing key developmental stages from early vegetative growth through senescence. Ten flights with the most similarity in timing were selected from each year for this study. Radiometric calibration was performed for multispectral imagery using in-field reflectance panels. Vegetation indices derived from UAS imagery included NDVI, NDRE, and ExG, representing canopy vigor, chlorophyll content, and greenness, respectively (Table 4) [7,15].

Manual phenotypic data was collected following the standard breeding protocol. Date to heading where 50% of heads in the plot are at stage Feekes 10.3 was recorded as days of the year past January 1 (DOY) [35] using the Field Book electronic tablet application [36]. To enable robust comparison across years and environments, all UAS observations were aligned using DRH. DRH was calculated as:

$$\text{DRH} = \text{Flight Date} - \text{Heading Date} \quad (1)$$

This alignment normalized temporal variation caused by differences in planting date and environmental conditions, ensuring that vegetation indices corresponded to comparable physiological growth stages across years.

Grain yield and test weight were collected using Mirus HarvestMaster GrainGage software (Juniper Systems, Logan, UT) integrated into a Wintersteiger Classic small plot combine (Wintersteiger, Innkreis, Austria). Histograms (Fig. 1) show the distribution of wheat yields for each growing season (2018–2021). In each subplot, the dashed vertical line marks the mean yield for that year, and the kernel density curve provides a smoothed representation of the yield distribution. The Fig. summarizes the range and variability of yield observations used in the study across multiple years. Environmental conditions varied substantially across years, providing a robust dataset for model development and evaluation [37]. These conditions included optimal precipitation in 2019, terminal drought in 2018 and 2020, and average rainfall in 2021.

Table 1
Flow diagram of the Texas A&M wheat breeding program.

Year	Trial Location and Name	Generation	Plots	Reps
1	Greenhouse Crossing Block		500	
2	Greenhouse rows	F_1	200	
3	Field plots	F_2	300	1
4	Field plots	F_3	500	1
5	Field plots	F_4	450	1
6	Head-rows	$F_{4:5}$	20,000	1
7	Preliminary Observation yield trials	$F_{4:6}$	600	1
8	Year 1: Advanced yield trials	Preliminary	200	2
9	Year 2: Advanced yield trials	Advanced	120	2
10	Texas Elite (TXE)		40	3
11	Southern Regional Performance Nursery (SRPN), TXE, and increase		50	3
12	SRPN, TXE, Uniform Variety Trial (UVT), increase, and Wheat Quality Council (WQC)		40	3
13	Release		1	

2.3. UAS data processing

Raw image data was processed using established workflows Fig. 2 [7]. In summary: SlantView software (SlantRange, Inc., San Diego, CA, USA) was used to process raw multispectral (MS) sensor images to acquire radiometrically calibrated MS images. These images, along with raw RGB images, were processed using Agisoft Metashape software (Agisoft LLC, St. Petersburg, Russia, 191144) to generate geospatial data products such as a 3D point cloud, Digital Surface Model (DSM), and orthomosaic images. A plot boundary polygon layer was overlaid on the orthomosaic and spectral values within each polygon were extracted to calculate ExG, NDVI, and NDRE.

2.4. Modeling and explainable AI framework

The overall workflow of the proposed framework is presented in Fig. 3. The framework integrates multi-source agricultural data, machine learning-based yield prediction, and temporal feature interpretation to identify biologically meaningful growth stages and optimal UAS sensing periods for wheat yield estimation. The workflow begins by integrating UAS-derived spectral reflectance indices collected across multiple growth stages with agronomic, environmental, irrigation, and genotype-related variables. Following data integration, preprocessing steps, including missing-value imputation, categorical encoding, and feature standardization, were applied to ensure data consistency and compatibility across models. Multiple machine learning algorithms representing linear, ensemble, boosting, and neural network approaches were then trained and optimized using repeated random subsampling and Bayesian hyperparameter optimization. Model predictions were evaluated across iterations to ensure robust estimation of predictive performance and generalization. Ensemble-based prediction strategies were also incorporated to improve model robustness and reduce model-specific variability.

To improve interpretability, an explainable AI framework was implemented to extract model-specific feature importance values from each learning algorithm. Feature importance values were normalized and aggregated across iterations to assess feature stability and consistency. Because many spectral variables correspond to specific crop growth stages defined by DRH, feature importance values were mapped back to their associated temporal positions to generate temporal importance profiles. The resulting temporal interpretation enabled identification of critical growth stages and optimal UAS acquisition windows associated with maximum predictive relevance. This framework extends beyond conventional predictive modeling by linking machine learning interpretability with crop phenology and operational UAS management strategies, thereby supporting more efficient and biologically informed decision-making in precision agriculture.

2.4.1. Model selection

A diverse set of machine learning models was used to capture both linear and nonlinear relationships between input features and wheat yield. The modeling framework included linear regression methods (Ridge Regression and Elastic Net), a multivariate regression approach (Partial Least Squares Regression, PLSR), tree-based ensemble models (Random Forest and Extra Trees), a boosting algorithm (Extreme Gradient Boosting, XGBoost), and a fully connected neural network. These models were selected to span a broad range of statistical and machine learning paradigms, enabling robust predictive performance across varying data structures and feature interactions.

In addition to predictive capability, the multi-model framework was

Table 2
Agronomic information of 2018-2021 Bushland Irrigated and Bushland Trials.

	Dryland		Irrigation			
Year	Planting Date	Harvest Date	Planting Date	Harvest Date	Nitrogen (kg ha^{-1})	Irrigation (mm)
2018	October 11, 2017	June 14, 2018	October 17, 2017	June 8, 2018	22.4	343
2019	October 20, 2018	June 26, 2019	November 15, 2018	July 3, 2019	67.3	88
2020	October 15, 2019	June 19, 2020	October 22, 2019	June 16, 2020	112.1	157
2021	November 4, 2020	June 26, 2021	October 12, 2020	June 22, 2021	89.7	209

Table 3
Unoccupied aerial system operating specifications and sensors used in Bushland Dryland and Bushland Irrigated nurseries from 2018-2021.

Unoccupied Aerial System	Sensor	Altitude (m)	Overlap (%)	Spectra	Ground Sampling Distance (cm pixel^{-1})
DJI Phantom 4 Pro		20	85	B, G, R	0.548
DJI Matrice 100	SlantRange 3P	35	70	G, R, RE, NIR	0.02

GSD: ground sampling distance, B: blue, G: green, R: red, RE: red edge, NIR: near infrared

Table 4
Vegetation indices generated to characterize vegetative growth of wheat breeding nurseries at Bushland, TX from 2018-2021.

Spectral Reflectance Indices	Formula	Reference
Normalized Difference Vegetation Index (NDVI)	$\frac{NIR - R}{NIR + R}$	[16]
Normalized Difference Red Edge Index (NDRE)	$\frac{NIR - RE}{NIR + RE}$	[34]
Excess Greenness Index (ExG)	$ExG = 2G - (R + B)$	[31]

designed to support explainable AI analysis. Model-specific feature importance measures were extracted from each algorithm, enabling comparative assessment of feature relevance across models and across repeated training iterations. This approach enables both robust prediction and consistent interpretation of feature contributions, forming the basis for subsequent temporal and decision-support analysis.

2.4.2. Data preprocessing

Before model development and training, all input features were systematically preprocessed to ensure data quality, consistency, and unbiased model training. Missing values were imputed with the mean, preserving all observations while minimizing distortion of feature distributions. Categorical variables were transformed with one-hot encoding to enable integration into models. Continuous variables were standardized with z-score normalization to ensure comparable feature scaling across models, particularly for algorithms sensitive to feature magnitude.

To prevent data leakage and ensure valid model evaluation, all preprocessing steps, including imputation, encoding, and scaling, were performed exclusively on the training data within each iteration and then applied to the corresponding test data. This approach preserves the integrity of the evaluation process and reflects real-world deployment. After preprocessing, the feature matrix was constructed by excluding the target variable (yield) and the temporal identifier (year) and combining all processed numerical and encoded categorical features into a unified input representation suitable for model training.

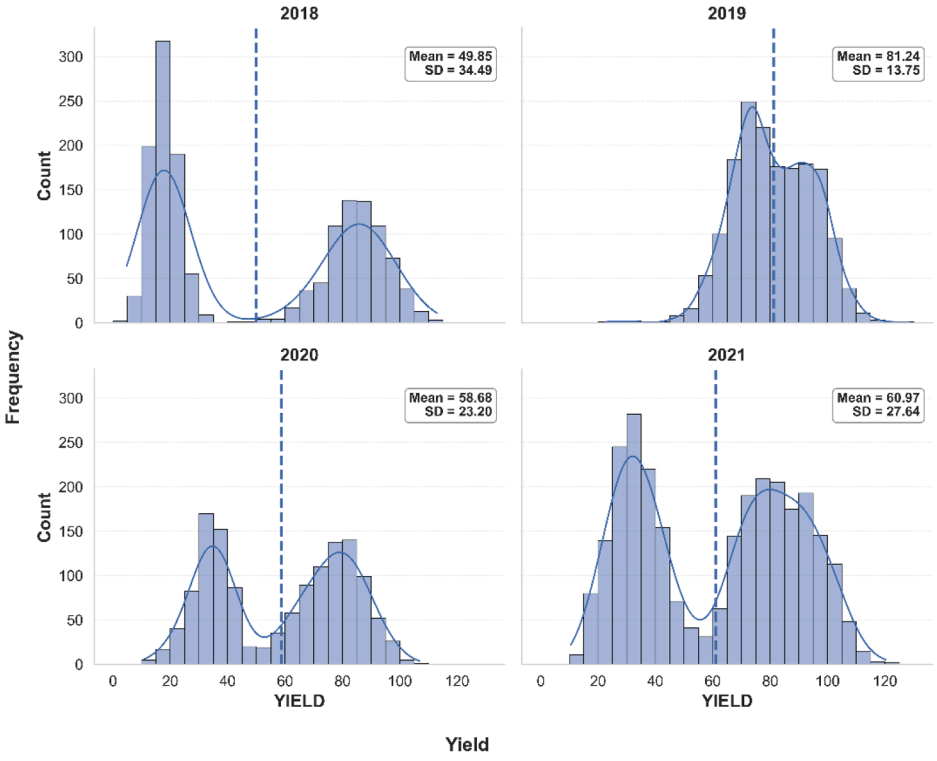


Fig. 1. Year-wise distribution of wheat yields across growing seasons.

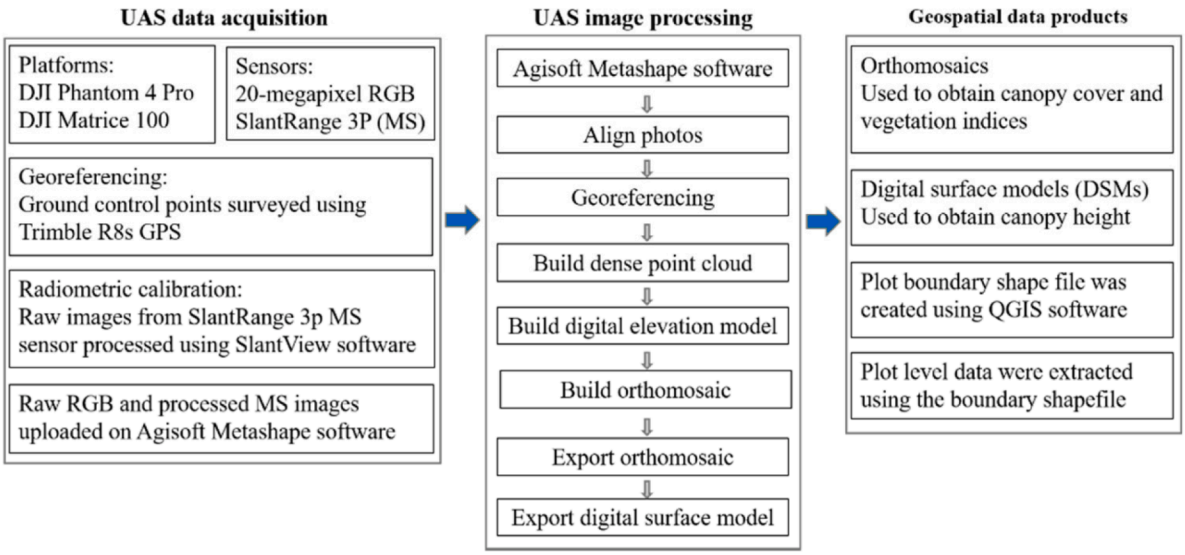


Fig. 2. UAS data collection and processing workflow.

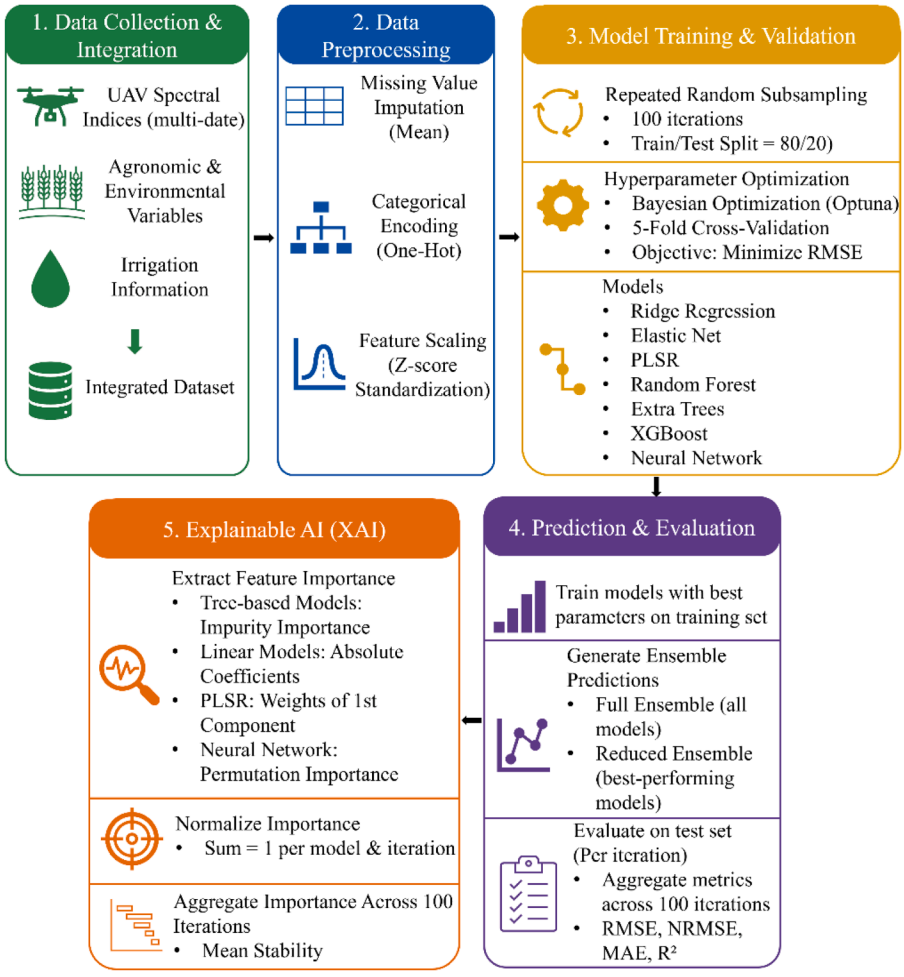


Fig. 3. Workflow of the proposed modeling framework for wheat yields prediction.

2.4.3. *Model training and validation*

A repeated random subsampling strategy was used to evaluate model performance and ensure robustness. In each iteration, the dataset was split into 80% training and 20% testing subsets, and the process was repeated 100 times. This repeated evaluation reduces sensitivity to a single data partition and provides a more reliable estimate of model generalization performance across varying data splits. Hyperparameter optimization was performed using Bayesian optimization in Optuna, with the objective of minimizing the root-mean-square error (RMSE). The hyperparameter search space for each model is provided in

Supplementary Table S1, while the final optimized hyperparameter values are summarized in Supplementary Table S2. During optimization, model performance was assessed using 5-fold cross-validation on the training data. This approach enables efficient exploration of the hyperparameter space while maintaining computational feasibility and avoiding overfitting to a single validation split.

Model-specific training procedures were applied to accommodate differences in learning behavior. Linear (Ridge, Elastic Net) and tree-based models (Random Forest, Extra Trees) were trained directly on the preprocessed training data using the optimized hyperparameters. For XGBoost, early stopping was implemented using a validation subset derived from the training data to prevent overfitting and determine the optimal number of boosting iterations. For the neural network model, early stopping based on validation loss was employed to improve generalization and prevent excessive training. Following hyperparameter optimization and model-specific tuning, each model was retrained using the optimal parameter configuration on the full training dataset. Additional implementation details, including software versions and the computational environment used for model development, are provided in Supplementary Table S3.

In addition to individual model predictions, an ensemble prediction was constructed by averaging outputs from multiple models to improve robustness and reduce model-specific bias. Two ensemble configurations were evaluated: (i) a full ensemble, incorporating predictions from all trained models, and (ii) a reduced ensemble, constructed using a subset of high-performing models identified based on their lower mean RMSE values. The final trained models were then evaluated on the corresponding test subset for each iteration, and performance metrics were aggregated across all repetitions to provide a robust assessment of predictive accuracy and model stability.

To assess whether differences in predictive performance among models were statistically significant, pairwise comparisons of RMSE values from the 100 repeated train–test splits were conducted using paired Wilcoxon signed-rank tests. The paired design accounts for the fact that all models were evaluated on identical data partitions in each repetition. To control the family-wise error rate arising from multiple pairwise comparisons, Holm's sequential correction was applied to the resulting p-values. Statistical significance was assessed at $\alpha = 0.05$.

2.4.4. Explainable AI framework

An XAI framework was implemented to interpret model predictions and quantify the contribution of individual features to wheat yield predictions. The framework consisted of two complementary components: (i) model-specific feature importance analysis to characterize feature contributions within each algorithm and (ii) a model-agnostic permutation importance analysis to independently validate the robustness of the identified feature importance patterns. Combining these two approaches provided both algorithm-specific interpretation and an independent assessment of feature relevance.

Feature importance was extracted using approaches appropriate to each model type. For tree-based models (Random Forest, Extra Trees, and XGBoost), impurity-based importance measures were used. For linear models (Ridge and Elastic Net), the absolute values of regression coefficients were used as indicators of feature contribution. For Partial Least Squares Regression (PLSR), the absolute weights of the first latent component were used to represent feature importance. These model-specific importance measures enable consistent interpretation across a diverse set of algorithms.

To address the known limitations of model-specific importance measures, including potential bias from correlated predictors and algorithm-specific characteristics, permutation importance was computed for each trained model. Permutation importance is a model-agnostic explainability method that estimates feature relevance by randomly permuting the values of a single feature in the held-out test dataset while leaving all other variables unchanged. The resulting decrease in predictive performance reflects that feature's contribution to model predictions. Because this approach directly evaluates changes in predictive accuracy on unseen data, it provides a complementary and more robust assessment of feature importance that is less dependent on the internal characteristics of individual algorithms.

For both analyses, feature importance values were normalized so their sum equals one within each model and iteration. Model-specific and permutation importance values were computed independently for each validation run and subsequently averaged to obtain stable estimates of feature importance. Many predictor variables represented spectral reflectance indices measured at specific days relative to heading (DRH). Feature importance values obtained from both the model-specific and permutation analyses were therefore mapped back to their corresponding DRH positions. Features representing the same temporal position were grouped to construct temporal importance profiles describing the evolution of predictor importance. Fig. 4 shows the workflow of the proposed explainable AI framework.

Overall, the proposed XAI framework extends beyond conventional model interpretation by integrating algorithm-specific explanations with an independent model-agnostic validation. This approach supports the identification of optimal UAS data acquisition windows, enabling more efficient and targeted data collection strategies in precision agriculture.

2.4.4.1. Temporal feature importance analysis. Temporal feature importance analysis was conducted to characterize how the predictive contribution of input features varies across crop growth stages. Each feature derived from spectral reflectance indices corresponds to a specific time point defined by DRH, enabling a direct link between feature importance and crop phenology.

For both model-specific feature importance and permutation importance analyses, normalized feature importance values were assigned to their corresponding DRH time points. Features measured at identical temporal position were grouped, and their values were aggregated to construct temporal importance profiles. This procedure was performed independently for every model and validation run before averaging to obtain stable temporal importance estimates.

To obtain stable temporal patterns, mean importance values were computed across iterations, and variability was assessed to evaluate consistency in feature contributions. Agreement between model-specific feature importance and permutation importance was interpreted as evidence that the temporal importance patterns were robust and not dependent on a particular explainability method.

This analysis transforms feature importance from a static ranking into a time-resolved signal, allowing identification of critical growth stages where spectral and agronomic features exert the greatest influence on yield prediction. The resulting temporal importance profiles provide a direct link between model interpretability and crop development dynamics.

2.4.4.2. Optimal UAS timing identification. Optimal UAS data acquisition timing was determined using temporal feature importance profiles from the XAI framework. Time periods with consistently high importance across models, iterations, and feature groups were identified as critical windows for data collection. To define these windows, temporal importance values were analyzed to detect peak regions and sustained high-importance intervals along the DRH timeline. Both magnitude (importance level) and consistency (frequency of occurrence across iterations and models) were considered to robustly identify informative time periods.

These windows mark stages when UAS-derived spectral reflectance indices contribute most to predictive accuracy. By prioritizing data collection during these periods, the framework enables more efficient UAS deployment, reduces redundant data acquisition, and enhances the practical applicability of yield prediction systems. This approach extends beyond traditional model evaluation by translating explainable AI

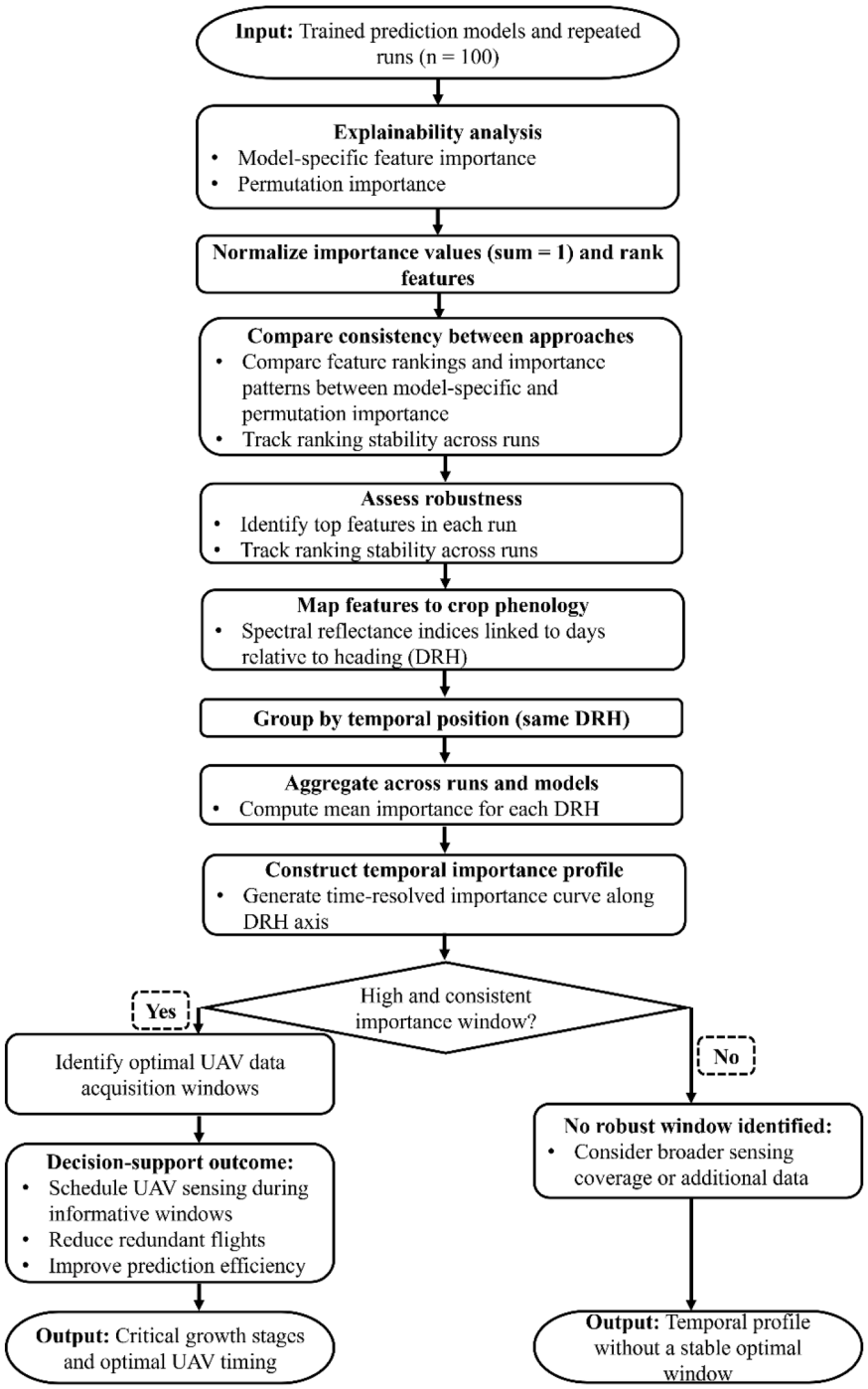


Fig. 4. Workflow of the proposed explainable AI framework.

outputs into actionable decision-support insights, specifically guiding when to conduct UAS-based sensing to maximize information gain.

3. Results

3.1. Model performance

Model performance was evaluated using the distribution of RMSE (Fig. 5). Among the evaluated models, tree-based ensemble methods, particularly XGBoost and Extra Trees, demonstrated the strongest predictive performance, achieving the lowest median RMSE (approximately 6.1) with tightly clustered distributions. This indicates both high accuracy and strong stability across repeated data partitions. Random Forest also performed competitively, though with slightly higher error than the top performing models.

Pairwise Wilcoxon signed-rank tests performed on the RMSE values from the 100 repeated runs confirmed that XGBoost and Extra Trees significantly outperformed Random Forest, Ridge, Elastic Net, PLSR, and the Neural Network (Holm-adjusted $p < 0.05$). However, no statistically significant difference was observed between XGBoost and Extra Trees (Holm-adjusted $p = 0.209$), indicating that both methods achieved comparable predictive performance.

In contrast, linear models (Ridge, Elastic Net, and PLSR) exhibited similar performance profiles, with median RMSE values clustered around 8.0–8.5. This suggests a limitation of linear approaches in capturing the underlying relationships in the dataset. The improved

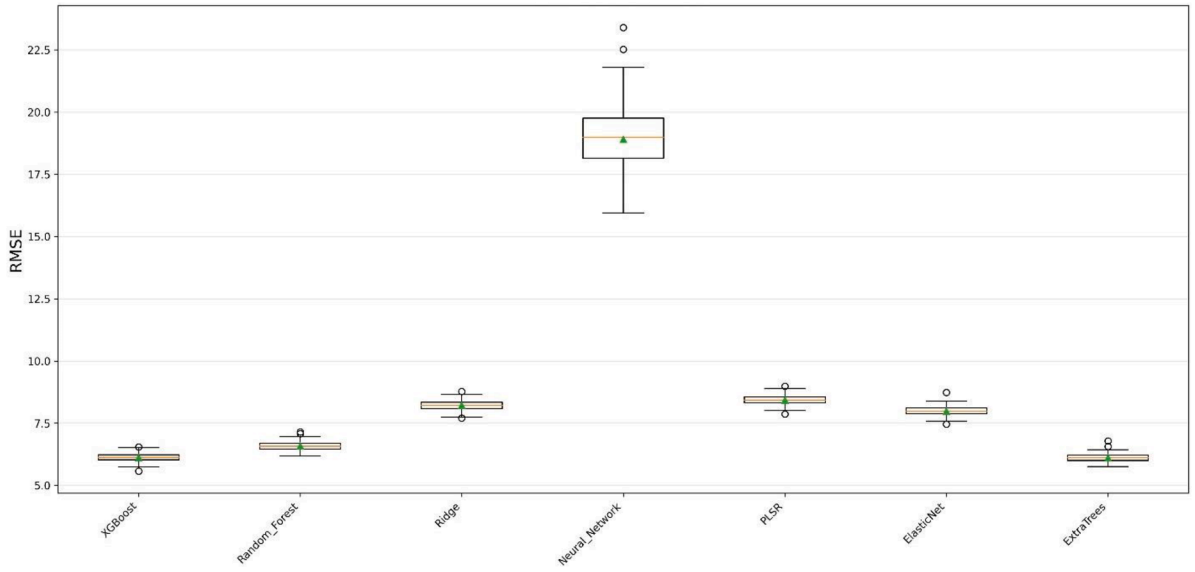


Fig. 5. Boxplot comparison of predictive performance (RMSE) for machine learning models.

performance of ensemble models indicates that the relationship between input features and wheat yield is likely nonlinear and complex, which these models are better suited to capture.

The neural network model showed substantially lower performance, with a median RMSE of approximately 19.0 and a much wider distribution. The larger interquartile range and the presence of high-value outliers indicate sensitivity to data partitioning and potential instability under the current configuration. This behavior suggests that the dataset characteristics or model setup may not favor deep learning approaches without further tuning or increased data volume.

Overall, most models demonstrated low RMSE variability, reflecting stable predictions across repeated runs. While minor outliers were observed across models, they remained within the distribution range, except for the neural network, which showed greater deviation.

Predicted versus observed yield relationships for all models are shown in Fig. 6. Overall, tree-based and ensemble approaches showed strong agreement with the 1:1 reference line, indicating high predictive accuracy. Among individual models, Extra Trees and XGBoost achieved the best performance, with RMSE values of 6.12 and 6.13 and R^2 values of 0.953 and 0.953, respectively. These models also produced the lowest mean absolute error (MAE) values (≈4.58–4.61) and the highest A10 scores (≈68.4–68.7%), indicating that a large proportion of predictions fall within 10% of the observed yield. The superior performance of these models reflects their ability to capture complex nonlinear relationships and feature interactions inherent in the dataset.

The ensemble approaches further enhanced predictive consistency.

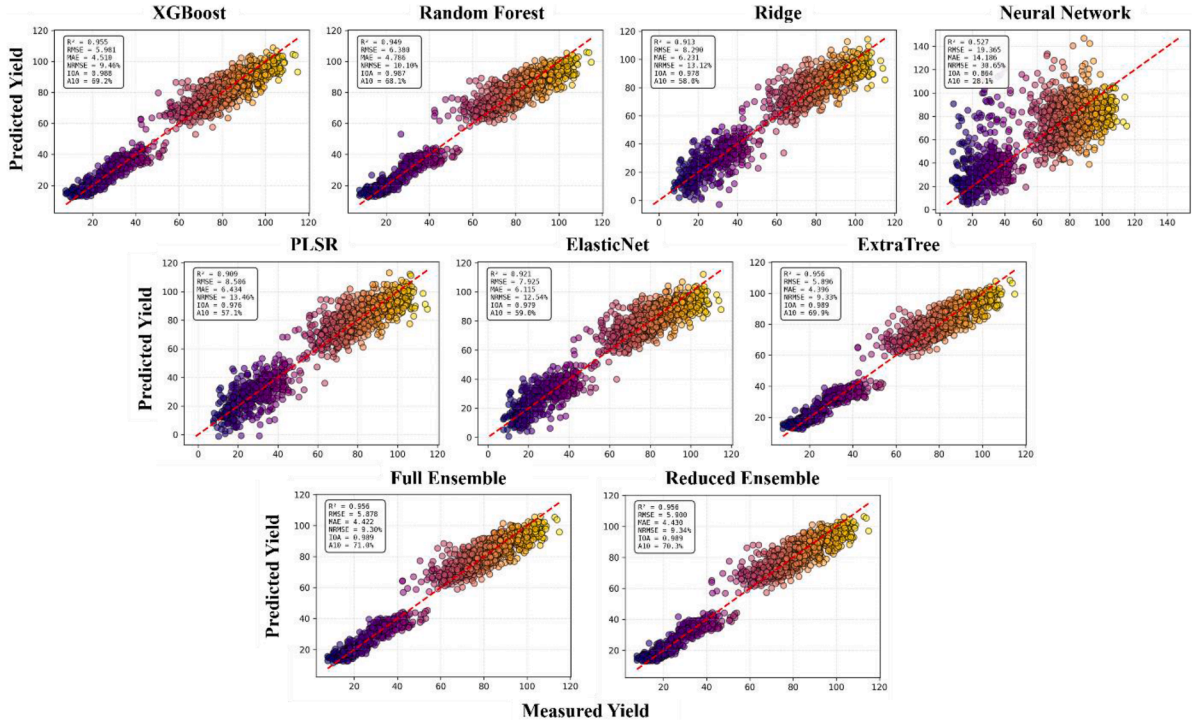


Fig. 6. Scatter plots of measured vs. predicted yield comparing individual machine learning models and ensemble (Full/Reduced) methods.

Both Ensemble Full and Ensemble Reduced exhibited tight clustering of predictions along the 1:1 line, indicating reduced variance compared to individual models. Notably, the reduced ensemble, constructed using the highest-performing tree-based models, achieved performance comparable to or slightly better than the full ensemble, with improved A10 values and similar error metrics. This result suggests that eliminating weaker-performing models from the ensemble reduces noise and improves the overall reliability of predictions. In particular, weighting models based on lower RMSE allows the ensemble to emphasize more accurate predictors, leading to enhanced performance without increasing model complexity.

Linear models (Ridge, Elastic Net, and PLSR) showed moderate performance, with RMSE values ranging from 7.99 to 8.44 and R^2 values between 0.910 and 0.920. While these models effectively captured the general trend in the data, the dispersion of points around the 1:1 line was more pronounced, particularly in the mid-yield range (40–80), indicating limitations in modeling nonlinear relationships.

In contrast, the neural network model demonstrated substantially lower performance, with a high RMSE (18.90), low R^2 (0.548), and the lowest A10 score (31.3%). The wide spread of predictions and higher variability suggest sensitivity to data structure and training conditions. This behavior may be attributed to the relatively limited dataset size and the tabular nature of the features, where tree-based ensemble models are generally more effective than deep learning approaches.

Overall, the results indicate that tree-based ensemble models provide the most accurate and stable predictions, while ensemble averaging further improves robustness. The statistical comparisons further confirm that XGBoost and Extra Trees form the highest-performing group, with no significant difference between them, while both significantly outperform the remaining individual models. The strong performance of the reduced ensemble highlights that a carefully selected subset of high-performing models can achieve equal or better predictive accuracy compared to a full ensemble. This finding is particularly important for subsequent analysis, as it supports the use of simplified and efficient model configurations without compromising performance, thereby providing a reliable foundation for explainable AI and temporal feature importance analysis.

3.2. Temporal correlation analysis of spectral features

Temporal analysis of spectral reflectance indices revealed strong, consistent relationships between crop phenology and yield prediction (Fig. 7). Evaluating correlations between vegetation indices and yield across DRH revealed clear temporal patterns, indicating that the predictive contribution of features varies substantially over the growing season.

Across all indices (avEG, avND, and avNR), correlations with yield increased steadily during the vegetative stage and peaked near the heading period. The avEG index showed the highest correlation at DRH 0, with a maximum |r| of 0.86, indicating strong sensitivity to crop conditions at the onset of the reproductive stage. In contrast, avND and avNR peaked slightly earlier, between DRH −15 and −10, with |r| values of 0.81 and 0.83, respectively. Despite these slight temporal offsets, all indices followed a consistent pattern, with predictive relevance rising rapidly around DRH −40, peaking near heading, and then gradually declining.

The way these feature values changed over time really backs up those findings. Mean index values rose consistently during the early and mid-growth stages, reflecting the development of the canopy and biomass, before leveling off or dipping slightly after heading. Meanwhile, the standard deviation for these indices increased as the growth stages progressed, which suggests that the crop conditions across different plots became more varied. This rise in variability is likely why we see that drop in correlation after the peak. It is important to note that the consistent trends across all indices highlight just how much the late vegetative and early reproductive stages influence yield. While avEG peaked right at heading, the earlier peaks for avND and avNR suggest they are more sensitive to what happens before heading, like chlorophyll buildup and the shaping of the canopy structure. These differences show that each spectral index picks up on different, complementary parts of crop growth, which together help make yield predictions much more reliable. Overall, the high correlation values (|r|>0.80) confirm that temporally aligned spectral features serve as strong predictors of yield. More importantly, this analysis demonstrates that predictive information is not uniformly distributed across the growing season. Instead, a critical temporal window exists between approximately -15 and 0 DRH, during which spectral features provide maximum predictive value.

3.3. Explainable AI-based feature importance analysis

The explainable AI framework revealed clear patterns in feature attribution across models, highlighting agronomic and temporal drivers of yield prediction (Fig. 8). Among all variables, irrigation and spectral indices near the heading stage consistently emerged as the most

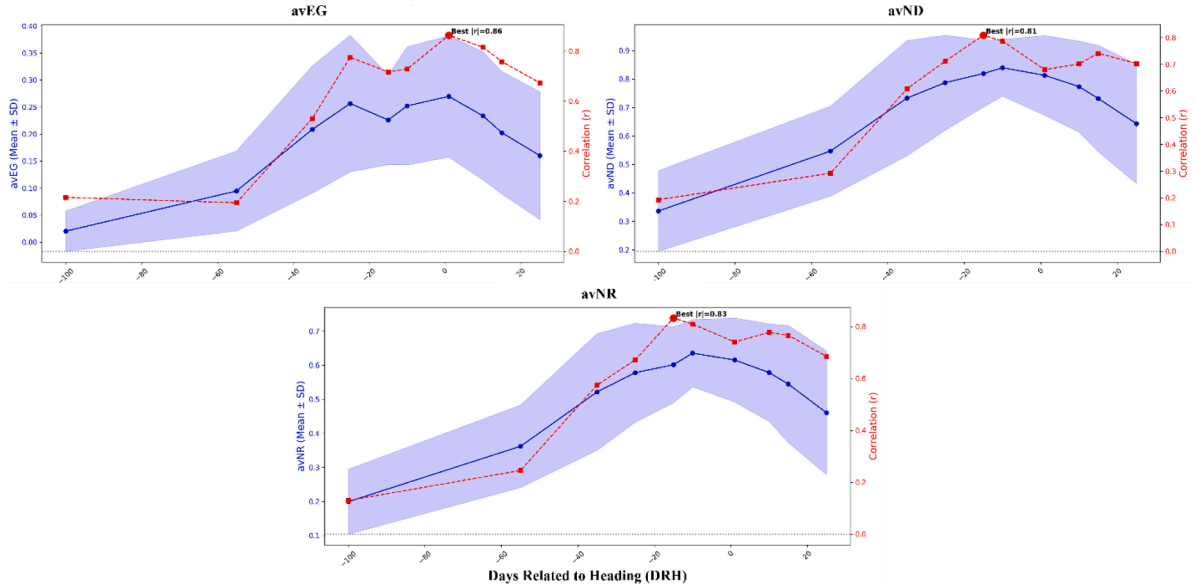


Fig. 7. Temporal profiles of spectral reflectance indices and their correlation with yield across days relative to heading (DRH).

influential predictors.

Across all the models we tested, irrigation stood out as a major factor, especially in the tree-based methods. This confirms that water management plays a dominant role in controlling yield variability. Alongside these management factors, our temporally aligned spectral features also pulled a lot of weight. Specifically, avEG at 1 DRH was the most influential feature for XGBoost, hitting a very high importance score of about 0.60. This really shows that canopy conditions right at the heading stage are absolutely critical for what the model predicts. Similarly, Random Forest leaned heavily on avNR at -15 DRH, which emphasizes how important those pre,heading signals are. We also saw some clear differences in how each model assigns importance. XGBoost was very focused, relying on just a few predictors to make its decisions. ExtraTrees, on the other hand, was more balanced, spreading the influence across irrigation and several different spectral features like avEG at 1 DRH and avEG at -25 DRH. These variations suggest that different models are actually picking up on different nuances of how the crop develops over time. Despite those differences, a consistent temporal pattern showed up across the board. Most of the high importance features were packed into the -25 to +1 DRH window. Anything outside of this range usually showed low importance, meaning those times didn't contribute much to the final prediction. This matches up perfectly with our temporal profiles of spectral indices, reinforcing our conclusion that the late vegetative to early reproductive stages hold the most critical information for estimating yield. In contrast, the linear models like Ridge, ElasticNet, and PLSR showed relatively low and flat importance values across all features, which just reflects their limited ability to catch the complex, nonlinear interactions that the tree-based models handle so well.

3.4. Validation of the explainable AI framework using permutation importance

To evaluate the robustness of the proposed explainable AI framework, permutation importance was computed as an independent, model-agnostic measure of feature relevance. Unlike model-specific importance metrics, permutation importance quantifies the contribution of each feature by measuring the reduction in predictive performance after randomly permuting its values in the held-out test data. This approach is less susceptible to biases associated with impurity-based feature importance and coefficient magnitude, particularly when predictors are correlated or differ in scale.

Fig. 9 compares the top fifteen features identified using permutation importance across the six machine learning models. Overall, the permutation-based rankings were highly consistent with those obtained from the model-specific feature importance analyses. Irrigation remained the most influential predictor for XGBoost, Random Forest, and Extra Trees, while vegetation indices collected near heading, particularly ExG at 25 days after heading (avEG_25_DRH), NDVI at 10 days after heading (avND_10_DRH), and ExG at 15 days after heading (avEG_15_DRH), were consistently identified as highly informative across multiple models.

Although the relative magnitudes of feature importance differed between the two explainability approaches, the dominant variables and temporal patterns remained largely unchanged. Features identified as highly important using the model-specific framework also received high permutation importance scores, indicating that the observed importance rankings were not artifacts of the specific importance metric employed.

Importantly, permutation importance confirmed that the greatest predictive information originated from spectral measurements collected during the late vegetative and early reproductive growth stages. This observation agrees with the temporal importance profiles (Section 3.3) and supports the identification of critical UAS sensing windows based on days relative to heading (DRH).

The strong agreement between model-specific feature importance and permutation importance provides independent evidence that the temporal importance patterns reported in this study are robust and not solely dependent on a particular explainability technique. Consequently, the proposed explainable AI framework provides reliable interpretation of feature relevance while maintaining computational efficiency for repeated analyses across multiple machine learning models and 100 independent repetitions.

3.5. Identification of optimal UAS data acquisition windows

Optimal UAS data acquisition windows were identified by integrating results from temporal correlation analysis (*Section 3.2*), model-based feature importance analysis (*Section 3.3*) and the independent permutation importance validation (*Section 3.4*). Together, these complementary analyses consistently demonstrated that the predictive relevance of spectral features is strongly time-dependent and concentrated within a narrow phenological interval surrounding heading.

Our temporal correlation analysis (Fig. 7) showed that vegetation indices like avEG, avND, and avNR hit their peak correlation with yield between -15 and 0 DRH. To be specific, avEG peaked right at DRH 0, while avND and avNR reached their max correlations a bit earlier, between -15 and -10 DRH. These results show that the shift from late vegetative growth to early reproductive development is easily the most informative stage for predicting final yield.

This is backed up by our feature importance analysis (Fig. 8), where

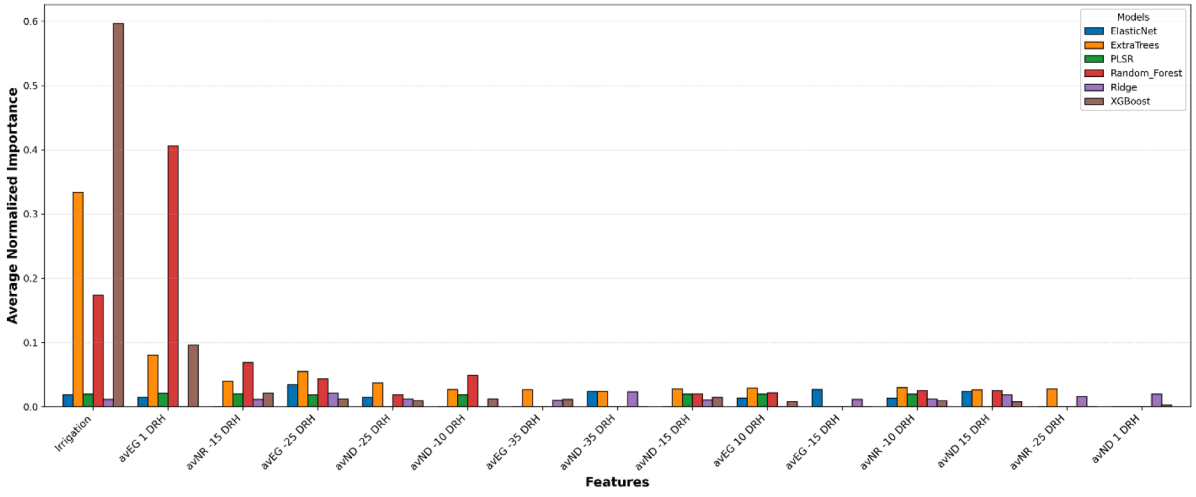


Fig. 8. Average normalized feature importance for the top 15 predictors across machine learning models.

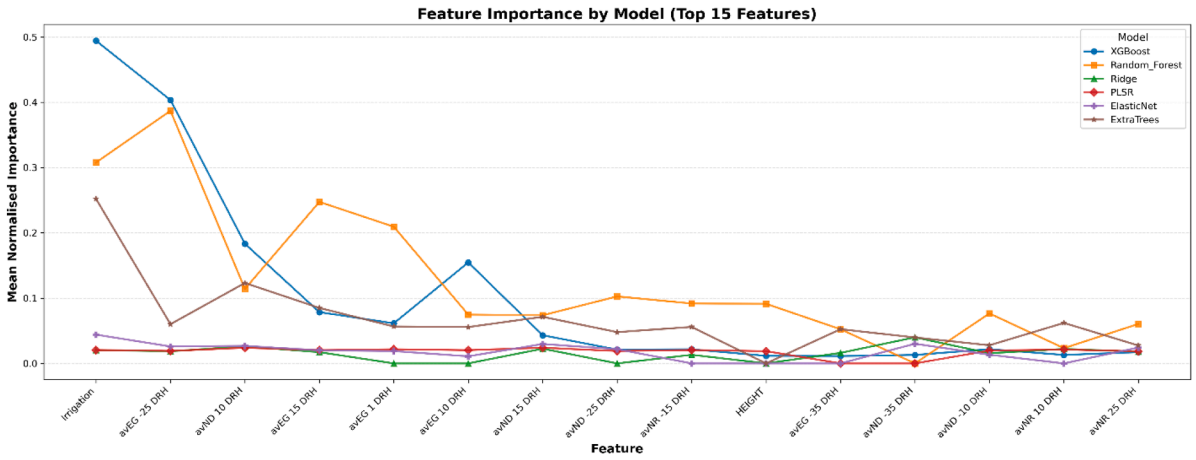


Fig. 9. Validation of the explainable AI framework using permutation importance across machine learning models.

the most influential predictors for the models were found in that same timeframe. Features like avEG at 1 DRH, avNR at -15 DRH, and avEG at -25 DRH were consistently at the top of the list, particularly for the tree-based models. Similar temporal patterns were also observed for the linear models and PLSR, although the relative importance magnitudes differed because of differences in model structure.

To further evaluate the robustness of the explainability framework, an independent permutation importance analysis was conducted (Fig. 9). Despite using a fundamentally different, model-agnostic approach based on changes in prediction error, permutation importance identified many of the same dominant variables, including irrigation and vegetation indices collected near heading. Although the relative importance values differed between the two explainability methods, the overall ranking of influential features and their temporal distribution were highly consistent.

Early growth stages (earlier than approximately -40 DRH) exhibited relatively weak correlations and low feature importance because canopy development was still limited. This is likely because of increased variability and the fact that canopy level indicators have less influence on yield as the season wraps up. Rather than conducting frequent or uniformly distributed flights throughout the growing season, data acquisition can be strategically concentrated within the identified- 15 to 0 DRH window to maximize information gain. Such targeted scheduling reduces unnecessary observations, improves operational efficiency, lowers data acquisition costs, and enhances the effectiveness of UAS-based crop monitoring systems. Overall, integrating temporal correlation analysis with two complementary explainability approaches enables the translation of machine learning predictions into actionable decision-support recommendations. The agreement between model-specific feature importance and permutation importance increases confidence in the identified critical sensing window and demonstrates that the recommended UAS acquisition period is robust across multiple analytical approaches. The identified optimal window therefore provides a data-driven foundation for scheduling UAS flights to capture the most informative stages of crop development for accurate yield prediction. Table 5 shows the summary for UAS data acquisition timing based on DRH-specific predictive relevance.

Table 5
Decision-support summary for UAS data acquisition timing based on DRH-specific predictive relevance.

Growth phase	DRH range	Predictive value	Recommended UAS action
Early vegetative	< -40 DRH	Low	Optional/background monitoring
Pre-heading transition	-40 to -15 DRH	Increasing	Useful if resources allow
Critical window	-15 to 0 DRH	Maximum	Prioritize UAS flights
Post-heading	> 0 DRH	Declining/ moderate	Secondary monitoring

3.6. Generalization of Unseen Data

The generalization capability of the machine learning models was evaluated using a Leave-One-Year-Out (LOYO) validation framework, in which each growing season was excluded once as an independent test set while the remaining years were used for training (Fig. 10). This strategy provides a more realistic assessment of model robustness under unseen environmental conditions than conventional random data splitting.

We noticed a lot of variation in how well the models performed from year to year, which showcases the environment impacts how well these models transfer. Overall, the 2019 season gave us the best results for cross-year generalization, with most models hitting lower error rates across the board. Things got much tougher when we used 2021 as the test year. The errors spiked across all models, suggesting that the weather, crop behavior, or management in 2021 was just fundamentally different from the other years, making it a much harder target to predict. When comparing the different approaches, the ensemble tree-based models generally handled these seasonal shifts better than the linear or neural network methods. XGBoost, Random Forest, and ExtraTrees were consistently competitive, proving they are more robust when it comes to dealing with seasonal changes and the complex, non,linear relationships in the spectral data. ExtraTrees and Random Forest, in particular, stayed relatively steady across the years, while linear models like Ridge, ElasticNet, and PLSR tended to swing wildly depending on which year was being tested. The Neural Network model showed inconsistent performance across unseen seasons. Although it achieved lower errors in some years, performance instability was observed under certain environmental conditions, suggesting sensitivity to temporal distribution shifts and limited robustness when trained on relatively constrained multi-year datasets. In contrast, ensemble-based methods appeared more capable of adapting to interannual variability and preserving predictive consistency. Fig. 11 shows temporal feature selection frequency across unseen test years and machine learning models under leave one year out (LOYO) validation.

Despite the observed year-to-year variability, all models retained reasonable predictive capability under LOYO validation, demonstrating that temporally aligned UAS-derived spectral features contain transferable information for wheat yield estimation across growing seasons. The increased prediction error observed for certain years further highlights the importance of incorporating diverse environmental conditions into

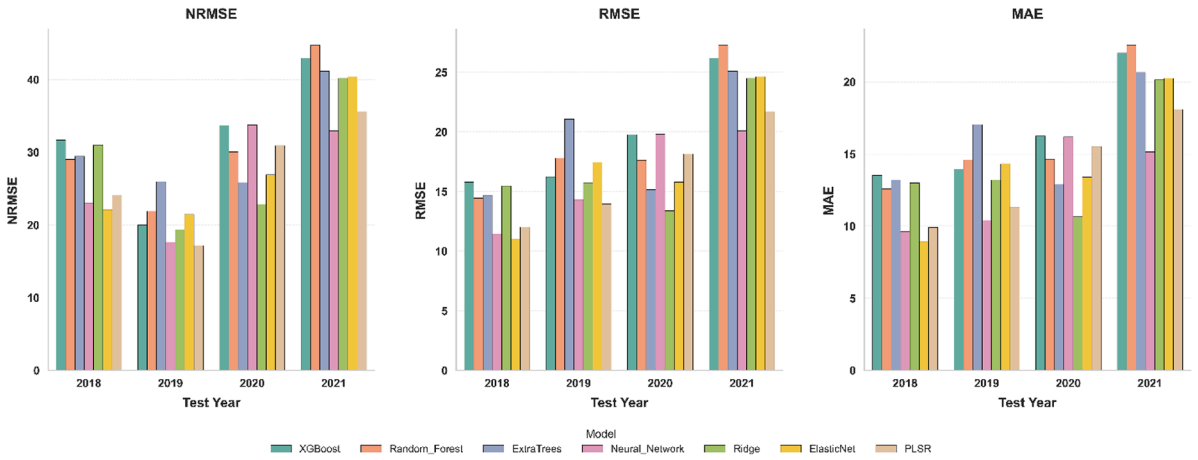


Fig. 10. Leave-One-Year-Out generalization performance across unseen growing seasons.

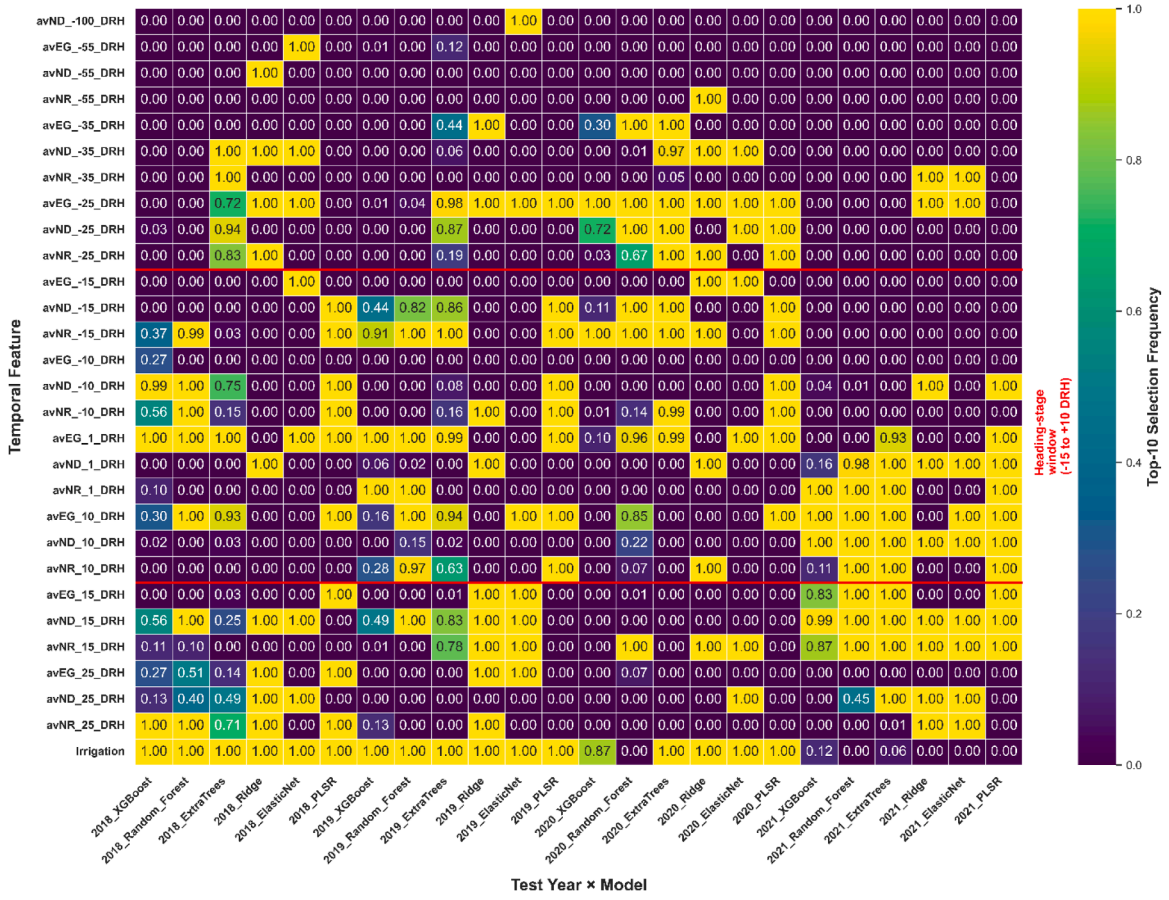


Fig. 11. Temporal feature selection frequency across unseen test years and machine learning models under leave one year out (LOYO) validation.

training datasets to improve model generalization and operational deployment potential.

To see how stable these features really were under unknown conditions, we used the LOYO feature attribution analysis. We tracked how often specific features were selected across different iterations to see which ones the models consistently relied on, regardless of which year or model was being used.

We noticed a clear pattern across different years, where the most important features were consistently clustered around the heading stage, specifically between -15 and +10 DRH. Within this window, certain features were repeatedly chosen as top performers across multiple models and even in test years the models hadn't seen before. This tells us that the heading and immediate post heading stages hold the most reliable information for predicting wheat yield. This trend held up even when weather and environmental conditions shifted between years, which really highlights how robust these heading stage spectral signals are. Looking at specific temporal features, avEG_1_DRH stood out as one of the most consistent predictors for almost every ensemble model and year we tested. We also saw high selection frequencies for features like avND_10_DRH, avNR_10_DRH, avND_15_DRH, and avNR_15_DRH. This just reinforces the idea that the late vegetative and early reproductive stages are the "sweet spot" for yield prediction. On the flip side,

early season data from -100 DRH or -55 DRH was rarely useful and showed very little consistency, suggesting that early canopy development just doesn't offer much predictive power. The models themselves actually behaved quite differently when picking features. Ensemble tree, based models, like XGBoost, Random Forest, and ExtraTrees, were very specific, consistently zeroing in on spectral features right within that heading stage window. This shows they are great at picking up on complex, nonlinear relationships tied to specific growth stages. Linear models, like Ridge, ElasticNet, and PLSR, were much broader and less focused, spreading importance across a wider timeframe. This suggests that linear approaches struggle to isolate those sharp, stage specific signals that are so critical for accuracy. While there was some year-to-year variation, likely due to different environmental stresses or shifting phenology, the dominance of the heading stage remained the one constant. Our LOYO (Leave One Year Out) analysis proves that predictive information isn't just scattered randomly throughout the season. Instead, there is a biologically meaningful and transferable window between -15 and +10 DRH where spectral data consistently generalizes across different years and models. These findings support a practical, operational strategy: if you want to maximize your predictive power while dealing with unpredictable environmental conditions, focus your drone flights on the heading stage interval.

4. Discussion

Currently, most yield studies lean on basic correlation analysis between spectral indices and the final harvest [18]. Our data confirms that peak correlations, especially for things like ExG right at heading (DRH 0), are great yield indicators. However, using an XAI framework shows us a much more detailed information landscape than a simple correlation ever could. The temporal importance patterns identified using the model-specific feature importance framework were further validated using model-agnostic permutation importance analysis. Despite differences in how feature importance is calculated, both approaches consistently identified irrigation and spectral reflectance indices collected near heading as the dominant predictors and highlighted the same critical phenological window for yield prediction.

Feature importance profiles showed that even as correlations start to decline during later reproductive stages, the informational content provided by certain variables remains relevant for capturing yield variability. Furthermore, our analysis identified critical pre-heading signals, such as NDRE and NDVI peaking between DRH -15 and -10, which provide complementary physiological data on chlorophyll and canopy structure that simple correlation at a single date might overlook. The use of feature importance highlights the value of early-stage monitoring in establishing a baseline for seasonal yield potential. Although correlations are naturally lower during early vegetative stages due to limited canopy development, these initial observations help the overall robustness of ensemble models by capturing early vigor and environmental response. By identifying a temporally stable window of high importance between −25 and +1 DRH that was consistently supported by both model-specific feature importance and permutation importance analyses. This approach ensures that data collection is concentrated during the transition to reproductive development, a period where canopy condition is most critical for yield formation, while maintaining the flexibility to incorporate early-season signals that improve the model's out-of-year generalization.

The concentration of predictive information during the period immediately preceding and surrounding heading is physiologically consistent with the processes that determine wheat yield potential. During stem elongation and the approach to heading, rapid canopy development and biomass accumulation increase leaf area and radiation interception, while canopy chlorophyll status reflects both photosynthetic capacity and plant nitrogen status. In this study, NDVI and NDRE were most strongly associated with yield approximately 10 -15 d before heading, consistent with these indices capturing differences among plots in canopy development, greenness, and chlorophyll status as the crop approached reproductive development. NDRE may be particularly informative during this period because red-edge reflectance remains sensitive to variation in canopy chlorophyll under relatively high leaf area, whereas NDVI becomes less responsive as dense wheat canopies approach saturation. ExG was most strongly associated with yield near heading, providing complementary information on visible canopy greenness. Thus, the convergence of NDVI, NDRE, and ExG within the broader heading-centered interval suggests that the XAI models were responding to multiple, complementary aspects of canopy condition rather than to an arbitrary temporal feature.

The biological importance of this period also reflects the transition from vegetative growth to reproductive yield formation. In wheat, the several weeks preceding anthesis include rapid spike growth and floret development, during which assimilate availability and crop growth strongly influence floret survival and ultimately grain number. Canopies capable of maintaining greater green leaf area, chlorophyll content, biomass accumulation, and radiation interception during this period have greater capacity to support both spike growth and photosynthesis as the crop enters reproduction. Because grain number is a major determinant of final wheat yield, spectral differences detected near heading may therefore integrate both the cumulative effects of earlier canopy development and the crop's capacity to establish reproductive sink strength. Following heading and anthesis, assimilate demand increasingly shifts toward developing grains, and maintenance of green leaf area and chlorophyll becomes important for sustaining radiation interception and photosynthetic supply during grain filling. This transition provides a physiological explanation for why spectral predictors retained importance immediately after heading, while the strongest and most consistent predictive information remained concentrated around the heading period.

Heading is also a particularly useful phenological reference from an operational breeding perspective. In addition to representing an important developmental transition associated with yield formation, heading date is a routinely recorded phenotypic trait in many wheat breeding programs because of its importance for adaptation, maturity, and evaluation of breeding material. Consequently, the phenological information required to express UAS observations as days relative to heading may already be available within breeding-program datasets and would not necessarily require collection of an additional specialized trait. This makes DRH alignment particularly amenable to implementation in breeding pipelines, where existing heading-date observations can be used to normalize UAS acquisition timing among genotypes and across growing seasons.

Importantly, the XAI results do not indicate that heading itself causes increased predictability or that heading represents a universally optimal sensing stage. Rather, phenological alignment appears to concentrate several physiologically meaningful sources of yield information within a common temporal reference. The repeated selection of heading-centered spectral features across model-specific and permutation-importance analyses indicates that the models consistently relied on canopy characteristics expressed during a period when source capacity, reproductive sink development, and final grain-number determination are closely coupled. This biological interpretation also explains why measurements taken very early in the season were less consistently informative: at those stages, canopy development is incomplete and much of the environmental and physiological variation that ultimately determines reproductive success and final yield has not yet been expressed. The heading-centered window identified by XAI should therefore be interpreted as the most consistently informative period within the environments and germplasm evaluated in this study, rather than as a universally optimal stage for wheat yield prediction.

The integration of XAI represents a significant shift from traditional black-box modeling toward a biologically grounded and operationally efficient phenotyping framework. While existing literature has predominantly focused on maximizing predictive accuracy, our results

demonstrate that feature importance analysis can be used to pinpoint the exact phenological windows that drive those predictions. By combining model-specific feature importance with model-agnostic permutation importance, we move beyond merely demonstrating predictive accuracy to providing robust and independently validated explanations of why the models perform well in the context of crop development. This transparency is crucial for wheat breeding programs, as it provides actionable guidance on when data acquisition is most critical, potentially reducing the need for redundant, high-frequency UAS flights that increase processing burdens without adding proportional predictive value.

5. Future work and shortcomings

This study was conducted using data collected from a single experimental station within one wheat breeding program over four growing seasons. Although an independent growing season (2021) was reserved for testing, the proposed framework was evaluated within the same environmental and management context. Consequently, the predictive performance and the identified optimal UAS acquisition window should not be interpreted as universally applicable across different climatic regions, management systems, or breeding populations. Additional evaluation across independent locations, breeding programs, and broader germplasm is needed to determine the generalizability of the proposed framework and its associated sensing recommendations.

The analysis also focused on a small set of UAS-derived vegetation indices, and future work could examine whether adding other sensor-derived traits or environmental variables improves prediction. Future studies should also evaluate the robustness of the proposed framework under different levels of simulated feature noise (e.g., sensor or measurement uncertainty) to assess the stability and reliability of model predictions under realistic operating conditions. Although temporal correlation and explainable AI analyses consistently identified the heading stage as the most informative period for yield prediction, this recommendation was inferred from model interpretation rather than directly evaluated through alternative flight scheduling strategies. Future studies should perform ablation analyses comparing predictive performance using (i) the complete set of UAS flights, (ii) only flights acquired near the heading stage, and (iii) flights collected before or after heading. Such analyses would directly quantify the trade-off between prediction accuracy and flight frequency and provide stronger evidence for optimizing UAS data acquisition while reducing operational costs.

Nevertheless, the proposed analytical framework, including phenology alignment using DRH, multi-model machine learning, and complementary explainable AI approaches is not inherently site-specific and can be readily applied to independent datasets collected under different environmental conditions. Future studies should evaluate whether the optimal sensing window identified here remains consistent across diverse production environments. A practical next step will be to test whether concentrating flights around this window can maintain prediction accuracy while reducing the cost and effort of repeated UAS campaigns in breeding programs.

6. Conclusion

This study demonstrated that phenology-aligned UAS-derived vegetation indices, combined with machine learning and explainable AI, can provide accurate and interpretable prediction of winter wheat grain yield across multiple growing seasons. Aligning spectral observations by DRH improved temporal consistency across years and hilighted biologically meaningful growth stages associated with grain yield. Tree-based ensemble approaches, particularly XGBoost, Extra Trees, and Random Forest, achieved the strongest predictive performance and the most stable generalization across seasons. Temporal correlation and feature importance analyses consistently showed that the most informative canopy measurements were concentrated near heading, with the highest predictive relevance occurring approximately between -15 and 0 DRH, with predictive relevance continuing slightly beyond this period. Furthermore, the temporal importance patterns identified using model-specific feature importance were independently validated using model-agnostic permutation importance. The strong agreement between these complementary explainability approaches confirmed that the identified key predictors and critical phenological window were robust and not dependent on a single feature importance method. These results indicate that UAS flights do not need to be uniformly distributed throughout the season; instead, strategically concentrating acquisitions around heading can improve efficiency while maintaining strong predictive performance. Overall, this framework advances wheat phenotyping by linking predictive accuracy with biological interpretability and operational decision support, providing a practical basis for more efficient UAS deployment in breeding applications.

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Statement on AI use

AI was used to assist with coding, formatting, and grammatical editing. The authors reviewed and edited the output as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Shannon Baker: Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation, Conceptualization. **Md Nafiul Islam:** Writing – original draft, Visualization, Software, Formal analysis. **Mahendra Bhandari:** Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Anjin Chang:** Writing – review & editing, Software, Methodology, Formal analysis, Data curation. **Amir MH Ibrahim:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition. **Jinha Jung:** Writing – review & editing, Supervision, Software, Resources, Project administration, Funding acquisition, Data curation. **Juan Landivar:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Shuyu Liu:** Writing – original draft, Resources. **Varenya Sri Mudumba:** Investigation, Data curation. **Ritik Pokharel:** Writing – review & editing, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Jackie Rudd:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition. **Jose Landivar Scott:** Methodology, Formal analysis, Data curation. **Thanos Gentimis:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.atech.2026.102606.

Data availability

Data will be made available on request.

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